

Few-Shot Law Acquisition: What Transfers Across Physics Laws, Measured

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TABLE I

FEW-SHOT LAW ACQUISITION (25% BUDGET) AND THE ABLATION DECOMPOSITION. THE VOCABULARY IS THE DOMINANT CARRIER; THE PHYSICS RESIDUAL CONSTRAINS TRAJECTORY SPECIALIZATION AT TINY BUDGETS.

Condition	median rel-MAE
Generalist (real signature)	0.1971046675358058
Dummy control	0.23896641936098253
From-scratch specialist	0.5803608226154111
Ablation: no vocabulary	2.021912348943419
Ablation: no physics	0.21398293623972417
Ablation: frozen body	0.19808596781154184

Abstract—Few-shot transfer in physics ML usually means: same equation, more parameters. We transfer across laws: a generalist that has seen five laws adapts to a held-out sixth with a quarter of the data at $2.9\times$ lower error than a from-scratch specialist. Then we decompose why, with ablations that overturned the intended narrative: the quantity vocabulary is the dominant carrier (removing it is $10\times$ worse), and the physics residual – helpful in general – constrains trajectory specialization at tiny budgets (removing it improves trajectory error $7\times$ while answers stay flat). Both findings are per-seed, committed, and re-runnable.

Index Terms—few-shot learning; transfer learning; physics-informed machine learning; transformers

I. INTRODUCTION

What does a model that has learned several physics laws know that a from-scratch learner does not? This paper measures the transfer and then decomposes its mechanism with ablations, reporting the decomposition as measured – including the direction that surprised the authors.

II. PROTOCOL

Five training laws, one held-out law, budgets from 24 to 96 samples. Conditions: generalist with real equation signature, dummy-signature control, from-scratch specialist. Ablations: no vocabulary stream, no physics residual, frozen body.

III. RESULTS

At the 25% budget the generalist is $2.9\times$ better than the specialist. Removing the vocabulary is catastrophic ($10\times$); removing the physics residual frees trajectory specialization $7\times$ while answers stay flat – the residual constrains curves at tiny budgets before data can specialize them.

REFERENCES

- [1] T. Brown et al., “Language models are few-shot learners,” NeurIPS 2020.